

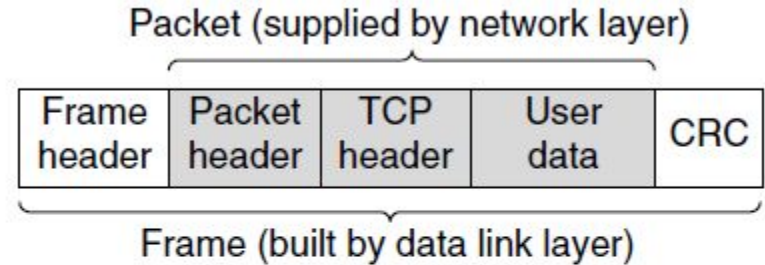
Network Layer

Er. Sudarshan Devkota

Network Layer

Application layer	Application gateway
Transport layer	Transport gateway
Network layer	Router
Data link layer	Bridge, switch
Physical layer	Repeater, hub

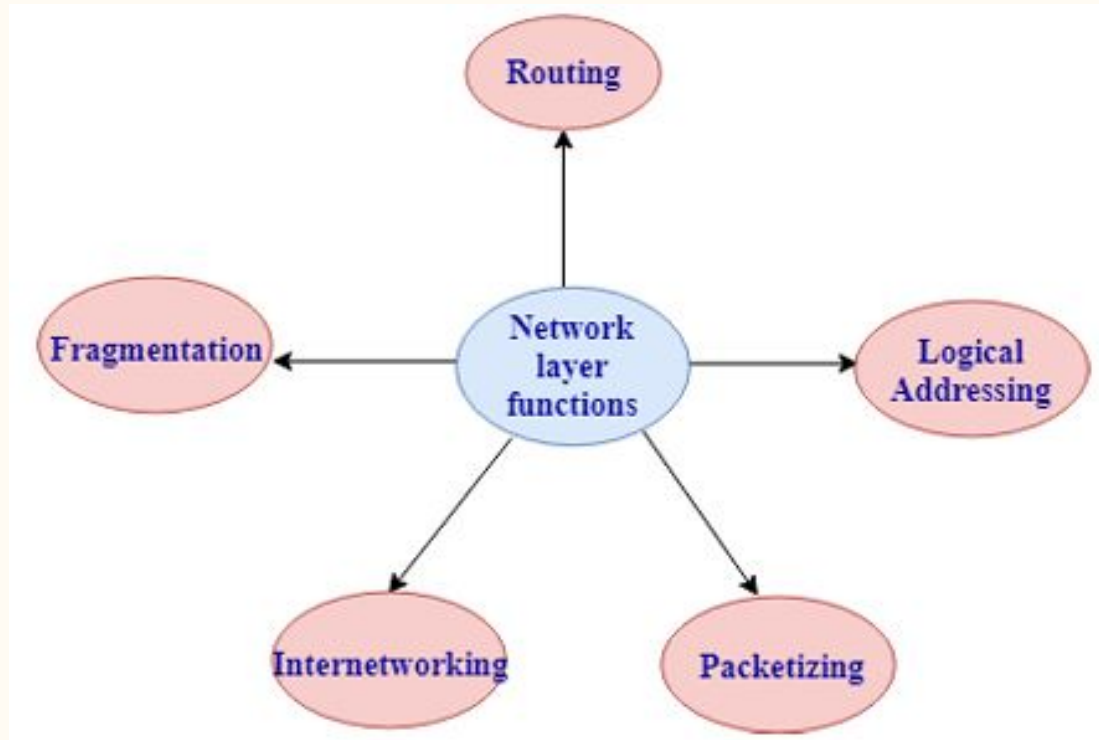
(a)



(b)

Figure 4-45. (a) Which device is in which layer. (b) Frames, packets, and headers.

Functions of Network Layer



Functions of Network Layer: Logical Addressing

- Every device is assigned a unique, software-defined logical address (such as an IP address), which functions as a universal identifier regardless of where the device is physically connected.
- Unlike physical MAC addresses which only work on a single local network segment, logical addresses are hierarchical and globally recognized.
- This system allows routers across the internet to inspect the destination IP label on a packet and know exactly which direction to forward it, effectively acting as a permanent digital mailing address.

Functions of Network Layer: Packetizing

- Data received from higher layers is often too large or unstructured for transmission, so the Network Layer encapsulates this data into standardized containers called packets.
- Each packet is wrapped with an IP Header, which serves as a "digital envelope" containing all the instructions necessary for intermediate systems to process the data.
- This header includes metadata such as the source and destination IP addresses, a Time-to-Live (TTL) counter to prevent packets from looping endlessly, and a protocol identifier to define the packet's contents.

Functions of Network Layer: Routing

- Routing is the intelligent, real-time process of choosing the most efficient path for a packet to traverse across a vast network topology.
- Routers maintain "Routing Tables," which act as maps of the internet; these tables are consulted every time a packet arrives to determine the best "next hop" toward its final destination.
- Because network conditions constantly fluctuate due to congestion or hardware failure, dynamic routing protocols (like OSPF or BGP) allow these tables to update automatically to steer traffic away from broken or slow links.

Functions of Network Layer: Internetworking

- The internet is a collection of diverse physical technologies—including fiber optics, satellite uplinks, and Wi-Fi—that do not natively speak the same language.
- The Network Layer acts as a universal translator, providing a standardized protocol that allows these disparate physical networks to function together as one cohesive, global system.
- Routers serve as gateways at the boundaries of these networks, receiving a packet from one technology and re-encapsulating it to be sent over another without the user experiencing any interruption in service.

Functions of Network Layer: Fragmentation

- Every network segment has a specific Maximum Transmission Unit (MTU), which is the absolute limit on the size of the data frame it can transport at once.
- If a packet is too large for a specific network segment it must cross, the Network Layer performs fragmentation, breaking the original packet into smaller, manageable pieces that can bypass the size restriction.
- Each piece is assigned a specific sequence number, allowing the final destination host to reassemble the scattered pieces back into the original, complete packet exactly as it was sent.

Virtual Circuit & Datagram Subnets

The Network Layer provides two primary models for moving data from a source to a destination. These models determine how the network manages paths, resources, and reliability.

1. Virtual Circuit Networks(Connection Oriented)
2. Datagram Subnets (Connectionless)

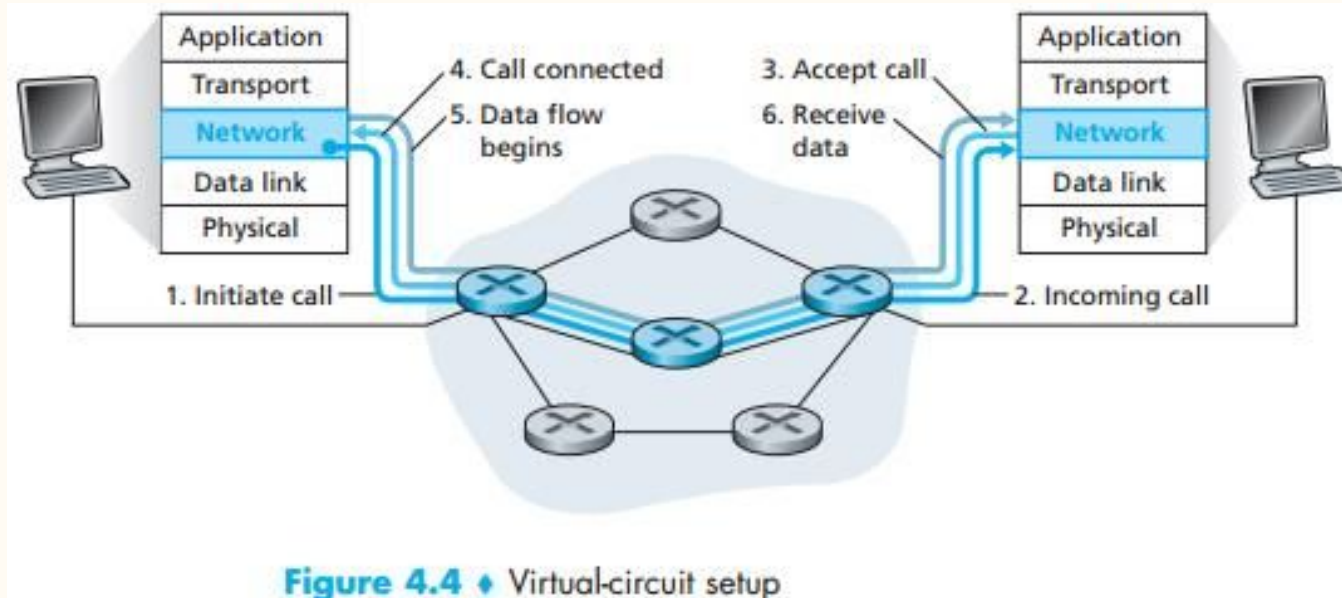
Virtual Circuit

Virtual Circuits (VCs) function like a formal telephone call. Before a single packet of data can be sent, a logical path must be explicitly established between the source and the destination host.

- The sender initiates a connection request that travels through the network.
- Every router along the intended path must acknowledge the request and verify that it has sufficient resources such as buffer capacity and CPU cycles to support the connection.
- Once the connection is established, all subsequent data packets follow the exact same physical path through the network.
- Because the path is already set, packets do not need to carry a full destination address. Instead, each packet contains a small, locally significant number called a Virtual Circuit Identifier (VCI).
- Each router along the path uses this VCI to look up exactly which outgoing port to use, making the forwarding process extremely fast.

Virtual Circuit

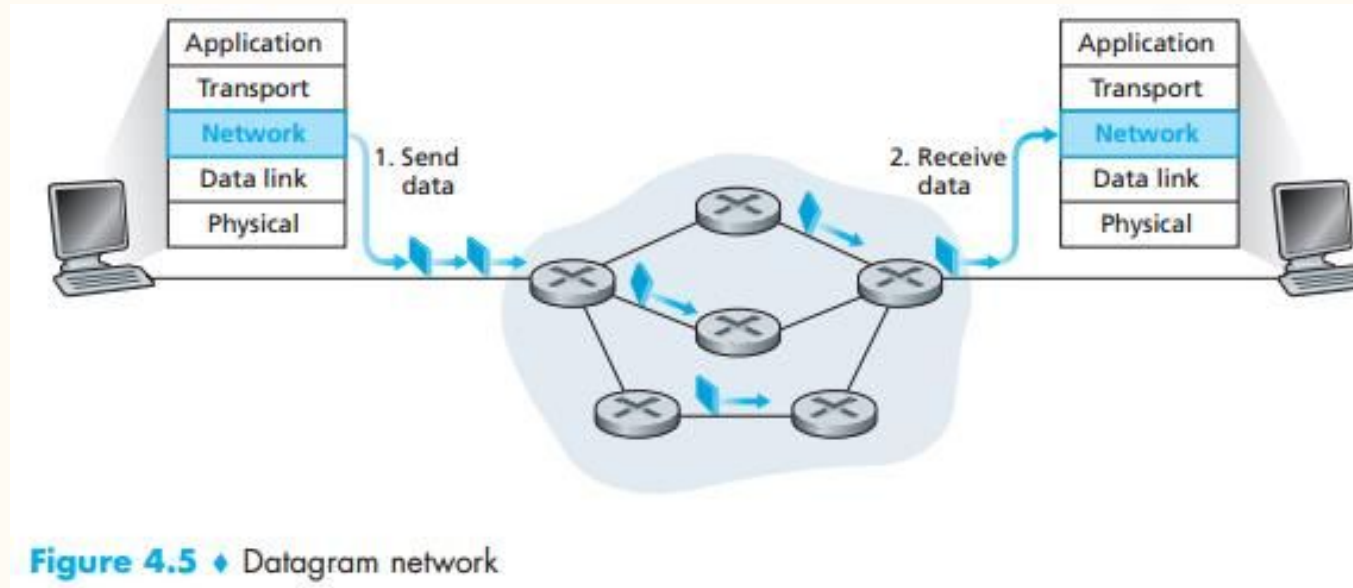
- Data arrives in the exact order it was sent.
- The connection must be manually closed to free up network resources.



Datagram

- Data transmission begins immediately without any prior connection setup.
- Each packet is treated as a completely separate unit of data.
- Every packet carries a full destination IP address in its header.
- Each router independently chooses the best path for every individual packet.
- Packets from the same stream may take different routes due to congestion.
- Packets can arrive in a different order than they were sent.
- Routers do not store information about ongoing connections or previous packets.
- The network can instantly route around failed hardware or broken links.

Datagram



IP (Internet Protocol)

The Internet Protocol (IP) is the primary network-layer protocol responsible for the addressing and routing of packets across interconnected networks.

It provides a universal communication standard that allows diverse network technologies—such as Ethernet, Wi-Fi, and Fiber—to exchange data by treating every device as a unique, addressable node.

IP is responsible for end-to-end delivery of data, moving packets from the source host to the destination host by determining the optimal path across the global internet.

IP is a connectionless, "best-effort" delivery protocol. It does not guarantee that packets will arrive in order or without errors; it relies on upper-layer protocols (like TCP) to ensure data integrity and reliability.

IP(Internet Protocol) V4

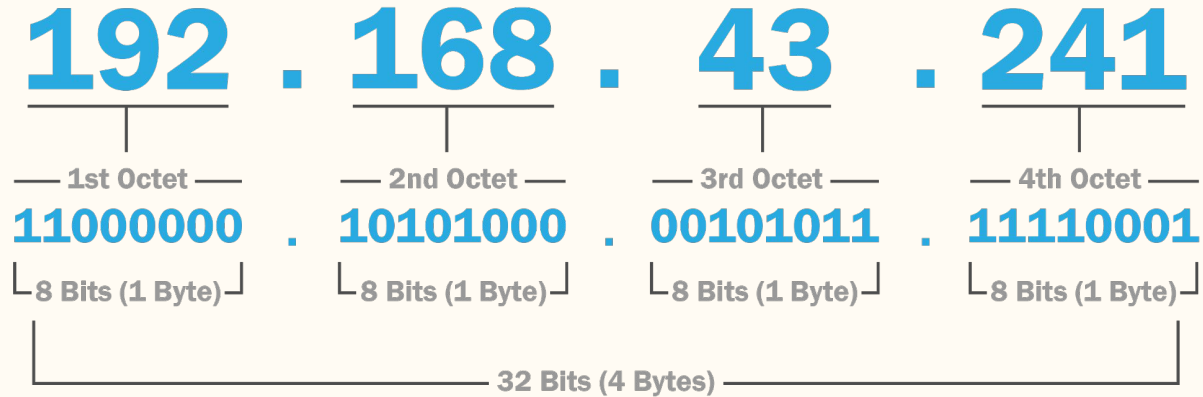
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IPV4 Address

- An IPv4 address consists of exactly 32 bits. This provides a total address space of 2^{32} , or approximately 4.3 billion unique identifiers.
- Computers represent these 32 bits as a sequence of ones and zeros (e.g.: 11000000.10101000.00000001.00000001).
- To make the binary manageable for humans, the 32 bits are divided into four 8-bit blocks known as octets. Each octet is converted from binary to a decimal integer (ranging from 0 to 255) and separated by periods (e.g., 192.168.1.1).
- Every address is divided into two parts:
 - **Network ID**
 - **Host ID**

IPV4 Address

IPv4 Address Format



IPV4 Address

Network ID: This identifies the specific "neighborhood" or segment of the network. Routers use this ID to determine the general destination of a packet. It is the common prefix shared by all devices on the same local network.

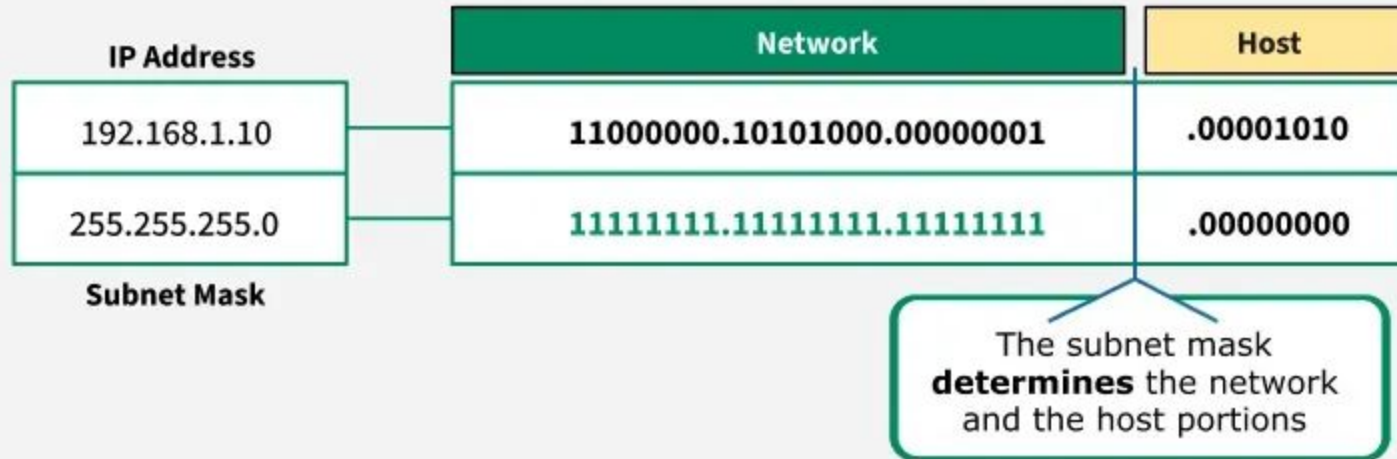
Host ID: This identifies the specific "house" or individual device within that network segment. Every device on a single network must have a unique Host ID to avoid communication conflicts.

Subnet: A subnet is a logical subdivision of an IP network. By creating subnets, you restrict broadcast traffic to a local segment, improve network security through traffic isolation, and simplify administrative management.

Subnet Mask: A Subnet Mask is a 32-bit binary number that tells a device which portion of its IP address belongs to the "Network" and which belongs to the "Host."

IPv4 Address

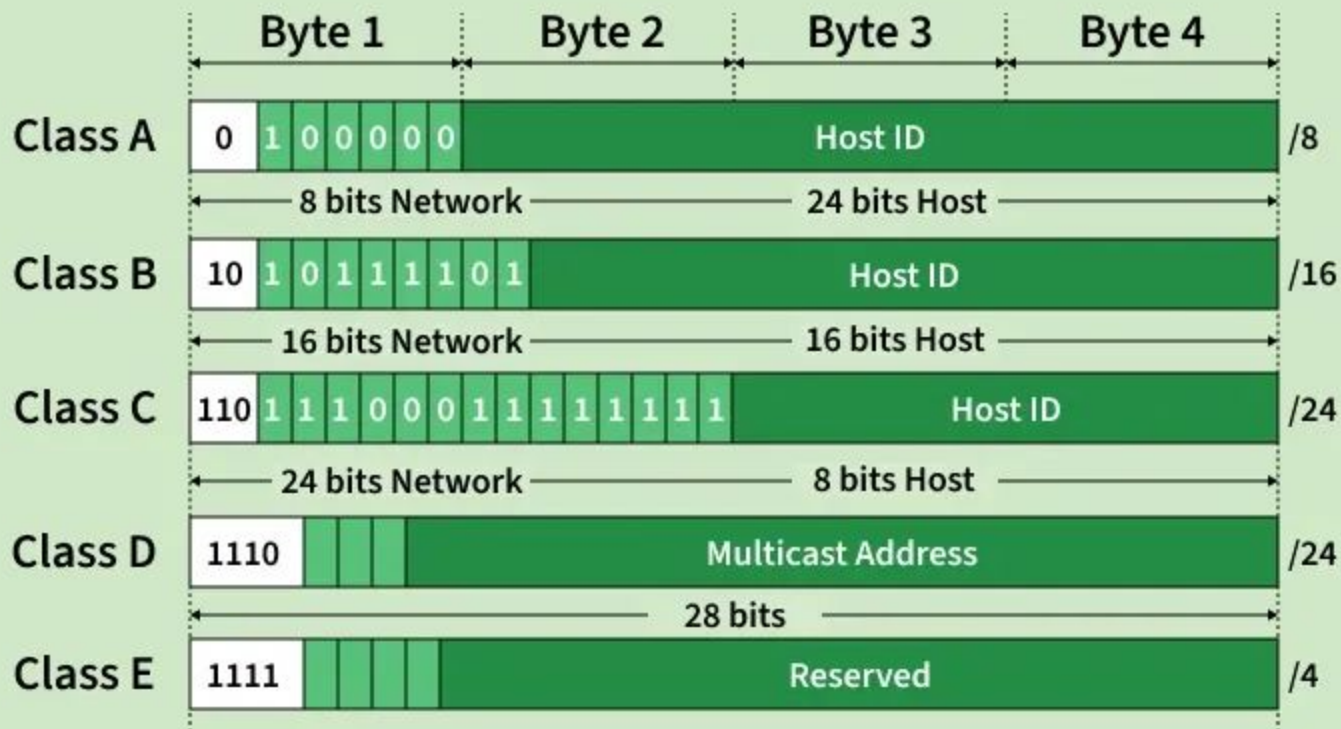
Role of Subnet Mask



Classes of IPV4 Address

Class	Left-most bit	Starting IP address	Last IP address
A	0xxx	0.0.0.0	127.255.255.255
B	10xx	128.0.0.0	191.255.255.255
C	110x	192.0.0.0	223.255.255.255
D	1110	224.0.0.0	239.255.255.255
E	1111	240.0.0.0	255.255.255.255

Classes of IPV4 Address



Classes of IPV4 Address

Class A: The "Massive" Network

- *Leading Bits:* Starts with 0.
- *Range:* 1.0.0.0 to 127.255.255.255.
- *Structure:* Network.Host.Host.Host (First 8 bits are the Network ID).
- *Capacity:* 126 possible networks; 16,777,214 hosts per network.
- *Use Case:* Designed for extremely large entities that require a massive number of devices on a single network, such as major government organizations or national telecommunication backbones.

Classes of IPV4 Address

Class B: The "Large" Network

- *Leading Bits:* Starts with 10.
- *Range:* 128.0.0.0 to 191.255.255.255.
- *Structure:* Network.Network.Host.Host (First 16 bits are the Network ID).
- *Capacity:* 6,384 possible networks; 65,534 hosts per network.
- *Use Case:* Designed for mid-to-large scale organizations, such as universities or large multinational corporations, where a single organization needs to manage tens of thousands of internal hosts.

Classes of IPV4 Address

Class C: The "Small" Network

- *Leading Bits:* Starts with 110.
- *Range:* 192.0.0.0 to 223.255.255.255
- *Structure:* Network.Network.Network.Host (First 24 bits are the Network ID).
- *Capacity:* 2,097,152 possible networks; 254 hosts per network.
- *Use Case:* Designed for small businesses and residential ISPs. This was the most common class for local area networks (LANs) where the number of devices is relatively low.

Classes of IPV4 Address

Class D: The "Multicast" Network

- *Leading Bits:* Starts with 1110.
- *Range:* 224.0.0.0 to 239.255.255.255
- *Structure:* No Network/Host split.
- *Capacity:* N/A
- *Use Case:* Reserved for Multicasting. Instead of sending data to one specific host (Unicast) or every host (Broadcast), data is sent to a specific "group" of interested devices. This is commonly used for streaming live media (like stock market feeds or video conferencing) to multiple subscribers simultaneously.

Classes of IPV4 Address

Class E: The "Experimental" Network

- *Leading Bits:* Starts with 1111.
- *Range:* 240.0.0.0 to 255.255.255.255.
- *Structure:* No Network/Host split.
- *Capacity:* N/A
- *Use Case:* Reserved for research, development, and future experimental purposes. These addresses are not routable on the public internet and have never been formally deployed for standard host identification.

Classless Address

Problem:

If an organization needed 500 hosts, a Class C (254 limit) was too small, forcing them to take a Class B (65,534 capacity). This wasted over 65,000 addresses.

As the internet grew, the rigid 8/16/24-bit boundaries could not support the diverse needs of modern organizations.

Solution:

Classless Inter-Domain Routing (CIDR) was introduced to replace this. It allows the boundary to move anywhere in the 32-bit string (e.g., a /26 or /27 mask), allowing engineers to "cut" the address block to the exact size required.

Classless Address

VLSM(Variable Length Subnet Mask) is a technique that allows network administrators to use different subnet masks for different subnets of the same classful network.

Instead of applying a "one-size-fits-all" subnet mask to an entire block, VLSM allows you to "cut" subnets to the exact size needed.

If you need a subnet for 10 hosts and another for 100 hosts, VLSM allows you to assign a /28 mask to the former and a /25 mask to the latter, preventing the waste inherent in fixed-length subnetting.

Subnetting

The Network ID is the first IP address within a specific subnet range, characterized by having all bits in the host portion set to binary 0. It serves as the formal identifier for the entire sub-network.

The Broadcast ID is the final IP address in a subnet range, where all bits in the host portion are set to binary 1. It is a reserved address used to communicate with all devices on a specific subnet simultaneously.

The Host ID is the specific bit sequence within an IP address that uniquely identifies an individual interface or node on a given sub-network. The length of the Host ID is determined by the subnet mask. Specifically, any bit position represented by a 0 in the subnet mask corresponds to the host portion of the address. For example, in a /24 mask (255.255.255.0), the final 8 bits are reserved for the Host ID.

Subnetting (Class C)

Binary	Decimal	CIDR
00000000=	0	/24
10000000=	128	/25
11000000=	192	/26
11100000=	224	/27
11110000=	240	/28
11111000=	248	/29
11111100=	252	/30

We can't use a /31 or /32 because we have to have at least 2 host bits for assigning IP addresses to hosts.

Subnetting(Class C)

1. How many subnets does the chosen subnet mask produce?
2. How many valid hosts per subnet are available?
3. What are the valid subnets?
4. What's the broadcast address of each subnet?
5. What are the valid hosts in each subnet?

Subnetting

Example 1:

Given IP address: 130.45.32.56

Subnet Mask: 255.255.0.0

What is the subnet address?

Solution:

IP address in binary:

10000010.00101101.00100000.00111000

Subnet Mask in binary:

11111111.11111111.00000000.00000000

After binary ANDing IP with mask, we get the subnet address, which will be

10000010.00101101.00000000.00000000

i.e., 130.45.0.0 is the subnet address.

Subnetting

A host is assigned the IP address **192.168.10.150** with a subnet mask of **255.255.255.224**. Determine the Network ID, the Broadcast ID and the total number of usable host addresses in this subnet.

IP (150) : 10010110

Mask(224): 11100000

AND(128) : 10000000

Network ID is 192.168.10.128.

Total IP in the subnet = $2^5 = 32$

Broadcast ID = 192.168.10.159

Usable Host = $2^5 - 2 = 30$

Subnetting

Example 3:

Given IP address 192.168.5.85/24. Determine the network and host part of this address.

Solution:

IP Address: 192.168.5.85

Subnet Mask: 255.255.255.0

In binary mode,

IP Address: 11000000.10101000.00000101.01010101

Subnet Mask: 11111111.11111111.11111111.00000000

So, here, the first 24 bits (First 3 octets) are network bits and the last 8 bits (Last octet) are the host bits.

ANDing IP address and Subnet mask we get a network address that is 192.168.5.0.

Broadcast address is the last address i.e., 192.168.5.255

Valid host is the number between the network address and broadcast address i.e., 192.168.5.1- 192.168.5.254.

Subnetting

Classwork:

Example 4:

Given IP address 10.128.240.50/30. Determine network, broadcast and valid host address.

Public IP

A globally unique 32-bit identifier assigned by the Internet Assigned Numbers Authority (IANA) or regional registries (RIRs) that is reachable and routable across the global Internet.

Serves as the unique identity of a network gateway or a public-facing server (such as a Web or DNS server)

Due to the IPv4 address exhaustion, public IP addresses are a finite resource. They are typically leased from an Internet Service Provider (ISP) and involve a cost.

Public routers maintain massive routing tables specifically to move packets between these unique global addresses.

Private IP

IP addresses reserved for internal use within a private network (LAN) as defined by RFC 1918. These addresses are strictly "non-routable" on the public Internet.

Reserved Ranges:

Class A: 10.0.0.0 to 10.255.255.255 (16.7 million addresses)

Class B: 172.16.0.0 to 172.31.255.255 (Approx. 1 million addresses)

Class C: 192.168.0.0 to 192.168.255.255 (65,536 addresses)

Network Address Translation

NAT is the technology that allows a site to use a set of private addresses for internal communication and a set of global internet addresses for communication with the rest of the world.

The basic idea behind NAT is to assign each company a single IP address for Internet traffic.

Within the company, every computer gets a unique IP address, which is used for routing internal traffic.

However, when a packet exits the company and goes to the ISP, an address translation takes place.

Network Address Translation

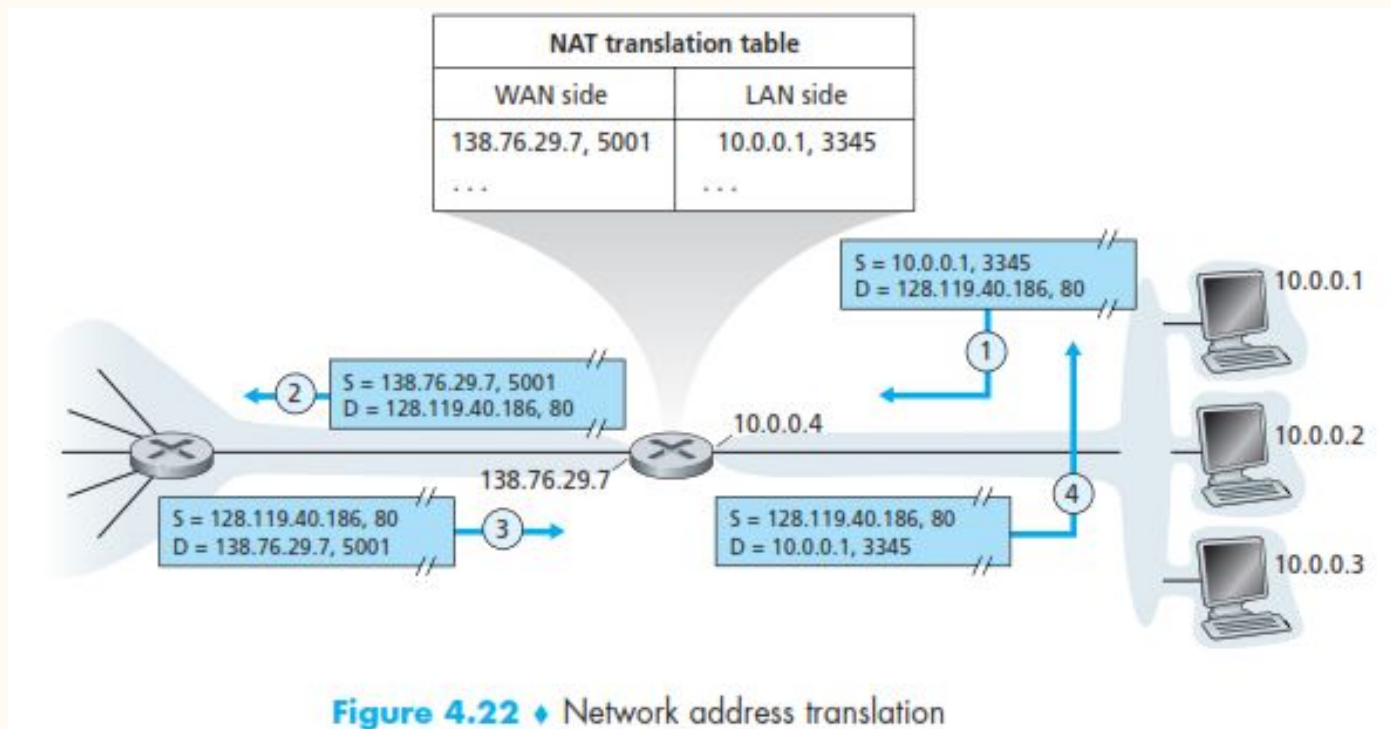
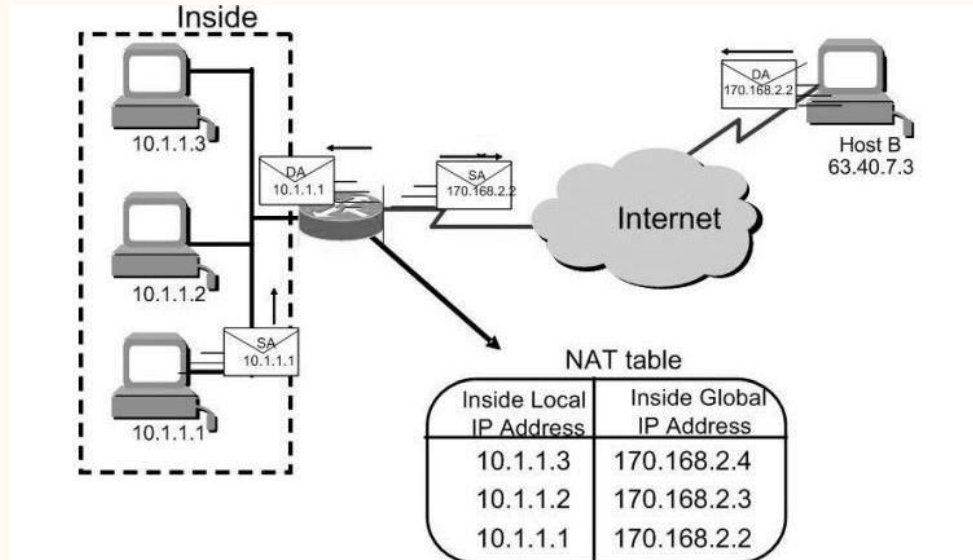


Figure 4.22 ♦ Network address translation

Types of NAT: Static

Maps one private IP to one specific public IP. Used typically for hosting internal servers that need to be accessible from the outside.



Types of NAT: Dynamic

A many-to-many mapping where private IP addresses are mapped to a "pool" of available public IP addresses on a first-come, first-served basis.

Used in corporate environments where several public IPs are available, and internal users need Internet access but do not need to host services.

Working:

When an internal host initiates a session, the router picks an unused public IP from its pre-configured pool and creates a temporary binding. Once the session ends and a timeout period passes, the public IP is returned to the pool for use by another host.

Types of NAT: Port Address Translation(PAT)/ NAT Overloading

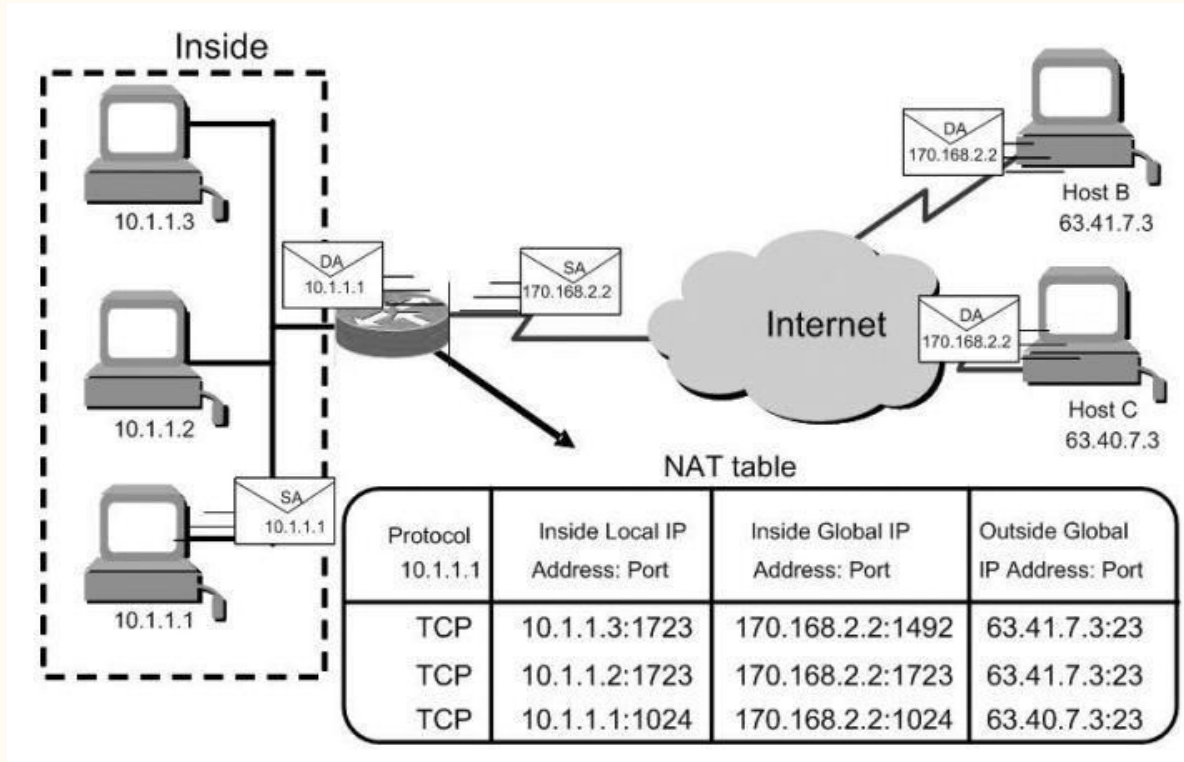
A many-to-one mapping that allows multiple internal private IP addresses to share a single public IP address by assigning unique source port numbers to each session.

The standard implementation for home networks and small-to-medium businesses where only one public IP is provided by the ISP.

Working:

The router tracks internal connections using the combination of Source IP + Source Port. For example, if two hosts (.10 and .11) both access a website, the router maps them both to the same public IP but gives them unique external ports (e.g., 203.0.113.5:10001 and 203.0.113.5:10002).

Types of NAT: Port Address Translation(PAT)/ NAT Overloading



NAT : Advantages

Address Conservation: Allows thousands of internal devices to share a single public IPv4 address.

Enhanced Security: Conceals internal IP addresses and network topology from external unauthorized users.

Network Flexibility: Permits changing internal IP schemes or ISPs without reconfiguring every local device.

Conflict Resolution: Enables multiple organizations to reuse the same private address ranges without global overlap.

Cost Reduction: Minimizes the need to purchase expensive, globally unique public IP addresses from ISPs.

NAT : Disadvantages

Performance Latency: Increases router overhead due to constant header modification and checksum recalculations.

Broken End-to-End Connectivity: Prevents external hosts from initiating direct sessions with internal devices.

Protocol Incompatibility: Disrupts applications that embed IP addresses in data payloads, such as VoIP or FTP.

Complex Troubleshooting: Makes tracing the original source of a packet difficult in centralized security logs.

P2P Communication Issues: Hinders direct peer-to-peer networking, often requiring complex "hole punching" techniques.

IPv4 Datagram Format

Version	IHL	Type of Service	Total Length	
Identification			Flags	Fragment Offset
Time to Live	Protocol		Header Checksum	
Source Address				
Destination Address				
IP Options (if any)				
Data				

IPv4 Datagram Format

The IPv4 header is a prefix added to every IP packet containing essential control information for routing and delivery. It has a variable length, with a minimum size of 20 bytes and a maximum of 60 bytes (if options are used).

Version (4 bits): Indicates the version of the IP protocol. For IPv4, this value is always set to binary 0100 (4).

Header Length (IHL - 4 bits): Specifies the length of the header in 32-bit words. The minimum value is 5 ($5 \times 4 = 20$ bytes), and the maximum is 15 ($15 \times 4 = 60$ bytes).

Type of Service (ToS/DSCP - 8 bits): Used for Quality of Service (QoS) management. It allows packets to be prioritized (e.g., low delay for VoIP or high throughput for large file transfers).

IPv4 Datagram Format

Total Length (16 bits): Defines the entire size of the packet (Header + Data) in bytes. The maximum possible size of an IPv4 packet is 65,535 bytes.

Identification (16 bits): A unique value assigned to a packet to help the receiver reassemble fragments of the same original packet.

Flags (3 bits): Controls fragmentation. The Don't Fragment (DF) bit prevents fragmentation, while the More Fragments (MF) bit indicates more fragments are coming.

Fragment Offset (13 bits): Indicates the exact position of the fragment in the original unfragmented IP packet, measured in units of 8 bytes.

IPv4 Datagram Format

Time to Live (TTL - 8 bits): A counter used to prevent packets from looping infinitely. Every time a packet passes through a router (a "hop"), the TTL is decremented by 1. If it reaches 0, the packet is discarded.

Protocol (8 bits): Identifies the next-level protocol in the data portion of the packet (e.g., 1 for ICMP, 6 for TCP, 17 for UDP).

Header Checksum (16 bits): Used for error-checking the header. If the checksum doesn't match at a router, the packet is discarded. It must be recalculated at every hop because the TTL changes.

Source and Destination IP Addresses (32 bits each): The IPv4 addresses of the original sender and the intended recipient.

Options and Padding: Used for network testing or debugging (like Record Route). Padding is added to ensure the header ends on a 32-bit boundary.

IPv4 Fragmentation

Every physical network technology (Ethernet, Wi-Fi, Token Ring) has a hard limit on the size of the frame it can carry, known as the MTU(Maximum Transmission Unit).

For standard Ethernet, the MTU is typically 1500 bytes. If a router receives a 4000-byte IP packet but needs to forward it over an Ethernet link, the packet is too large to fit into a single frame.

The router performs Fragmentation, breaking the large IP datagram into smaller pieces (fragments) so they can traverse the underlying physical medium.

IPv4 Fragmentation Process

Header Replication: When a packet is fragmented, the router creates multiple new packets. Each fragment receives a copy of the original IP header.

Data Partitioning: The original payload is sliced into pieces. Crucially, each piece (except the last one) must be a multiple of 8 bytes to accommodate the math of the Fragment Offset field.

Identification (16 bits): A unique number assigned to the original datagram. All fragments of the same original packet share the same Identification number so the receiver knows they belong together.

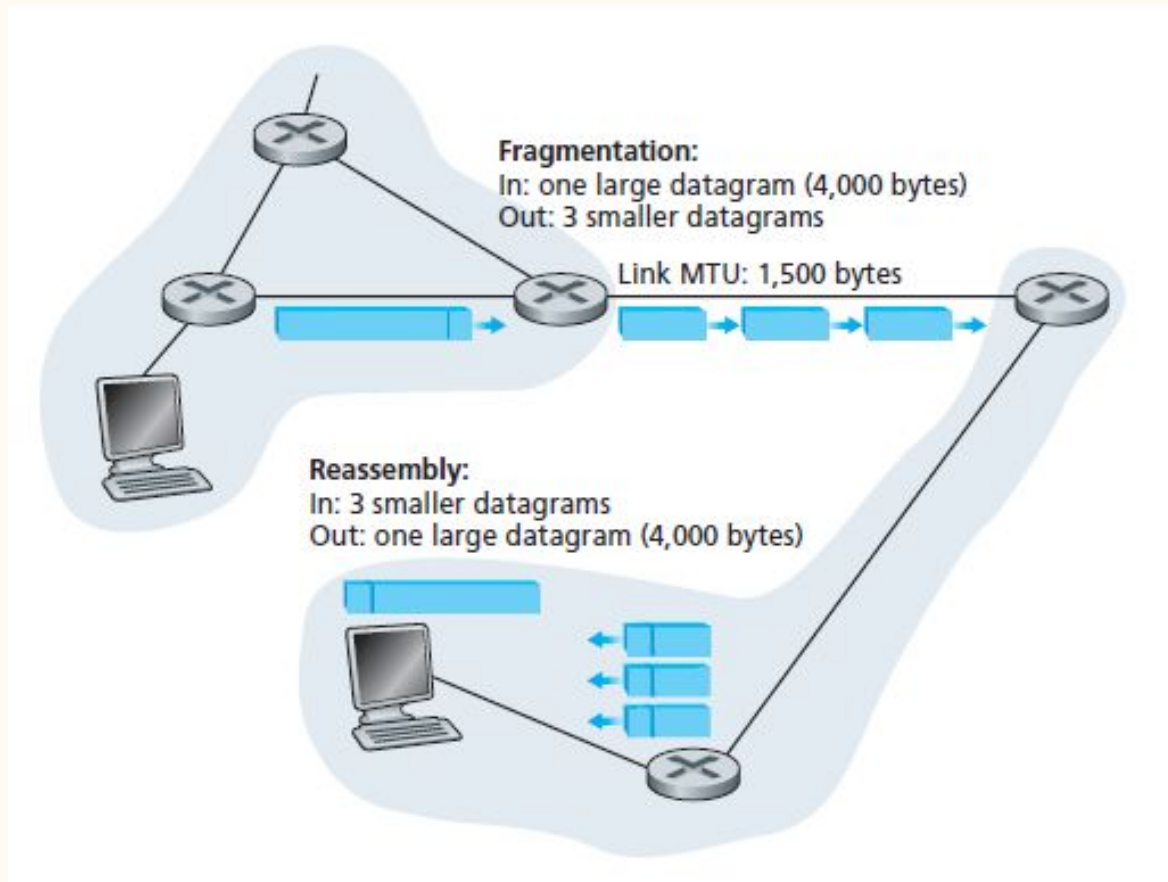
IPv4 Fragmentation Process

Flags (3 bits):

- *Reserved bit*: Always set to 0.
- *Don't Fragment (DF)*: If set to 1, the router is forbidden from fragmenting the packet. If the packet is too large, the router drops it and sends an ICMP error message back to the sender.
- *More Fragments (MF)*: Set to 1 for all fragments except the very last one. It tells the receiver, "There are more pieces coming."

Fragment Offset (13 bits): Acts as a pointer. it specifies where in the original datagram this particular fragment belongs. It is measured in units of 8-byte blocks.

IPv4 Fragmentation Process



IPv4 Fragmentation Process

Fragment	Bytes	ID	Offset	Flag
1st fragment	1,480 bytes in the data field of the IP datagram	identification = 777	offset = 0 (meaning the data should be inserted beginning at byte 0)	flag = 1 (meaning there is more)
2nd fragment	1,480 bytes of data	identification = 777	offset = 185 (meaning the data should be inserted beginning at byte 1,480. Note that $185 \cdot 8 = 1,480$)	flag = 1 (meaning there is more)
3rd fragment	1,020 bytes (= $3,980 - 1,480 - 1,480$) of data	identification = 777	offset = 370 (meaning the data should be inserted beginning at byte 2,960. Note that $370 \cdot 8 = 2,960$)	flag = 0 (meaning this is the last fragment)

IPv6

- IPv6 is the most recent version of the Internet Protocol, designed by the IETF to replace IPv4.
- It was developed to address the long-term problem of IPv4 address exhaustion and to provide a more efficient routing architecture.
- IPv6 uses a 128-bit address space, providing approximately 3.4×10^{38} unique addresses.
- This is a massive increase from the 2^{32} addresses available in IPv4.
- Addresses are written in Hexadecimal format, separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).

The "Sand" Analogy(Fun Fact):

If you were to assign an IPv6 address to every single grain of sand on Earth, you would still have enough addresses left over to repeat the process for every star in the observable universe and you would still only have scratched the surface of the available space. This is precisely why we no longer need NAT for address conservation

IPv6 Address Structure

IPv6 addresses are 128 bits in length, represented as 8 groups of 16 bits each. Each group is written as four hexadecimal digits (a "hextet") separated by colons.

Notation Rules for Simplification

Rule 1: Leading Zero Omission

- Within each 16-bit hextet, leading zeros are unnecessary for interpretation and can be removed.
- Example: 0DB8 becomes DB8; 0001 becomes 1.

Rule 2: Double Colon (::) Compression

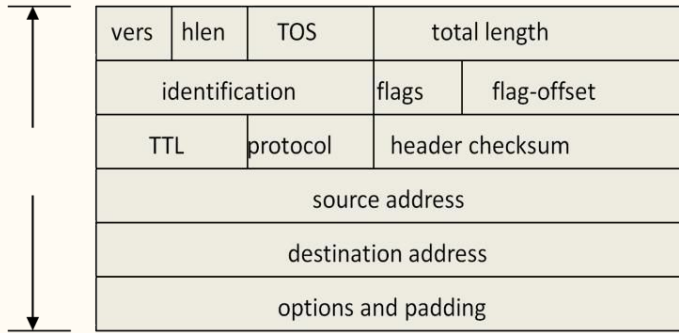
- A single contiguous string of one or more hextets consisting entirely of zeros can be replaced with ::.
- Constraint: This can only be applied once per address to prevent ambiguity when reconstructing the full 128-bit value.

IPv6 Address Structure

2001:0db8:3c4d:0012:0000:0000:1a2f:1234

Operation	Resulting Format
Remove leading zeros	2001:db8:3c4d:12:0:0:1a2f:1234
Apply Double Colon (::)	2001:db8:3c4d:12::1a2f:1234

IPv6 Header Comparision

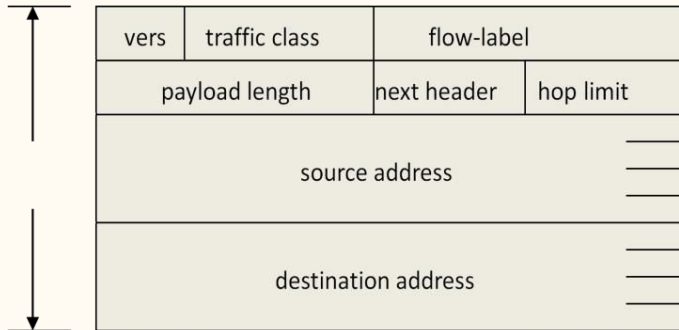


IPv4

Removed (6)

- ID, flags, flag offset
- TOS, hlen
- header checksum

Changed (3)



IPv6

Added (2)

- traffic class
- flow label

Expanded

- address 32 to 128 bits

IPV6 Header

0-3 Version	4-11 Traffic Class	12-31 Flow Label	
32-47 Payload Length		48-55 Version	56-63 Hop Limit
64-191 Source Address			
192-288 Destination Address			

IPv6 Header

Version (4 bits): Indicates the Internet Protocol version. For IPv6, the bit pattern is always 0110 (decimal 6).

Traffic Class (8 bits): Replaces the IPv4 "Type of Service" field. It is used by source nodes and routers to identify and distinguish between different classes or priorities of IPv6 packets (e.g., prioritizing real-time voice over bulk email).

Flow Label (20 bits): A new field used to tag sequences of packets that belong to a specific "flow" (e.g., a specific video stream). It allows routers to handle these packets through a specialized path without re-examining the entire header, facilitating efficient Quality of Service (QoS).

Payload Length (16 bits): Indicates the length of the data portion of the packet (including any Extension Headers) in octets. It supports a maximum payload of 65,535 bytes.

IPV6 Header

Next Header (8 bits): Identifies the type of header immediately following the IPv6 header. It uses the same protocol numbers as the IPv4 "Protocol" field (e.g., TCP = 6, UDP = 17) or identifies an IPv6 Extension Header.

Hop Limit (8 bits): Replaces the IPv4 "Time to Live" (TTL) field. It specifies the maximum number of router hops the packet can traverse. Each router decrements this value by 1; if it reaches zero, the packet is discarded.

Source Address (128 bits): The IPv6 address of the original originator of the packet.

Destination Address (128 bits): The IPv6 address of the intended recipient. In the case of Source Routing, this may be an intermediate node.

IPv6 over IPv4

Massive Address Space: With 2^{128} total addresses, IPv6 effectively eliminates the need for Network Address Translation (NAT) and allows for true end-to-end global connectivity.

Fixed Header Size: IPv6 uses a streamlined, fixed 40-byte header, which is faster for routers to process than the variable-length IPv4 header.

Checksum Removal: IPv6 removes the header checksum (relying on Layer 2 and Layer 4 checksums), reducing the CPU load on intermediate routers during every hop.

Improved Autoconfiguration (SLAAC): Through Stateless Address Autoconfiguration (SLAAC), devices can automatically generate their own unique IPv6 addresses upon connecting to a network, significantly reducing administrative overhead compared to DHCP in IPv4.

Mandatory Security (IPsec): While IPsec was an "add-on" for IPv4, it is a core, mandatory requirement for the IPv6 protocol stack, ensuring native encryption and authentication.

IPV6 over IPv4

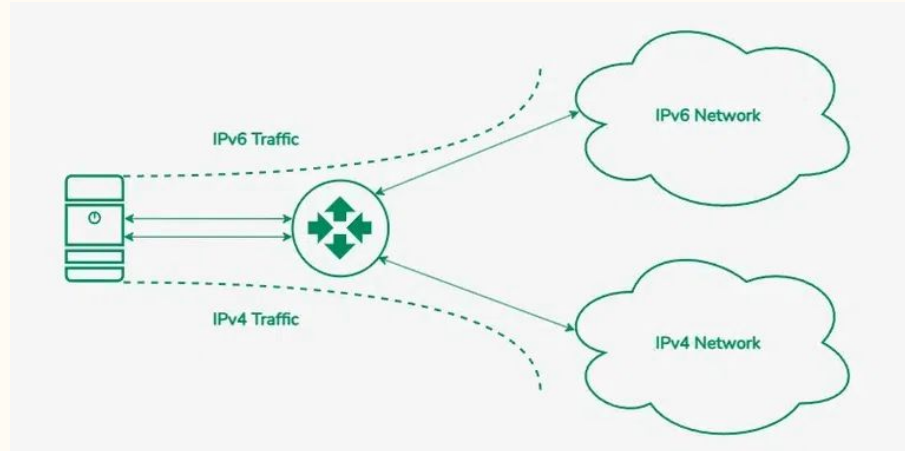
No Router-Level Fragmentation: IPv6 routers do not perform fragmentation. If a packet is too large, the source is notified via ICMPv6, shifting the processing burden to the endpoints.

Enhanced Multicast: Replaces the inefficient "Broadcast" traffic of IPv4 with optimized multicast and anycast delivery methods.

Transition from IPv4 to IPv6

1. Dual-Stack: The Bilingual Approach

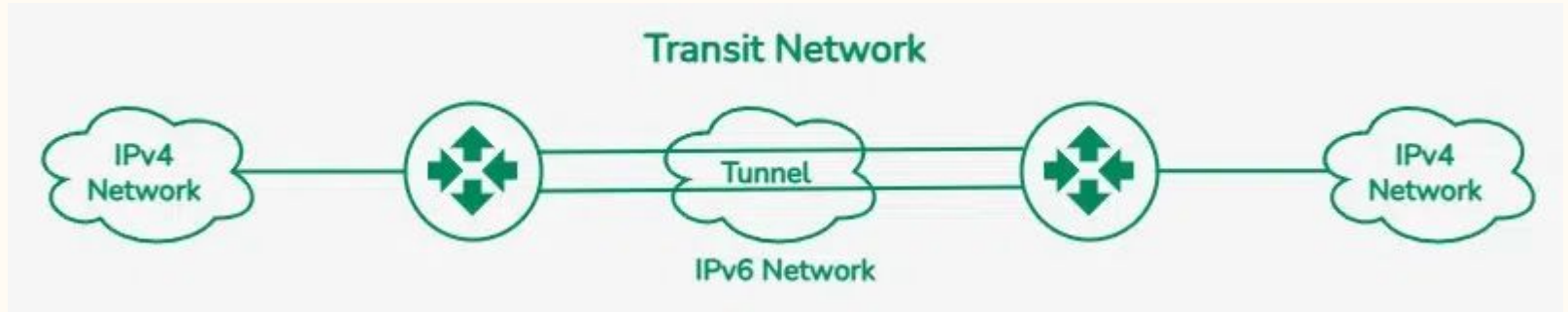
- This is the most common method. The device is simply configured to speak both IPv4 and IPv6 at the same time.
- Your computer and router are assigned both an IPv4 and an IPv6 address.
- If you visit a modern site (like Google), your computer uses its IPv6 "voice." If you visit an old site, it automatically switches to its IPv4 "voice."
- There is no translation; the device just picks the best path available.



Transition from IPv4 to IPv6

2. Tunneling: The "Envelope" Method

- Tunneling is used when you have two IPv6 networks that need to talk to each other, but the "internet" between them only understands IPv4.
- The router takes the IPv6 packet and wraps it inside an IPv4 envelope (header).
- The IPv4 network sees an IPv4 packet and moves it along. When it reaches the other side, the receiving router "unwraps" the envelope and delivers the original IPv6 packet.
- The middle of the journey doesn't even know it's carrying IPv6 data.



Transition from IPv4 to IPv6

3. Translation (NAT64): The Interpreter

- This is a "last resort" for when an IPv6-only device needs to talk to a strictly IPv4-only server.
- A gateway sits in the middle. It intercepts the IPv6 packet, strips off the IPv6 header, and physically replaces it with an IPv4 header.
- It works exactly like a human language translator. It changes the message format so the destination can understand it.
- This is the most "expensive" method in terms of processing power because the router has to rewrite every single packet.

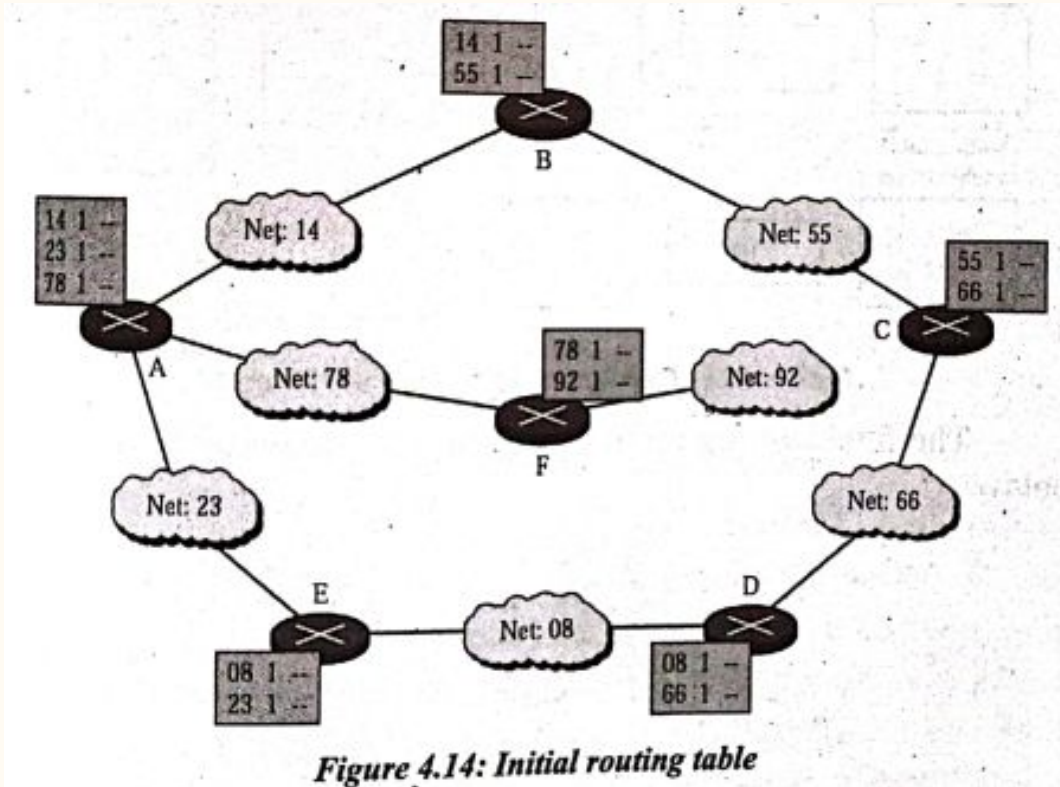


Routing Algorithms: Distance Vector

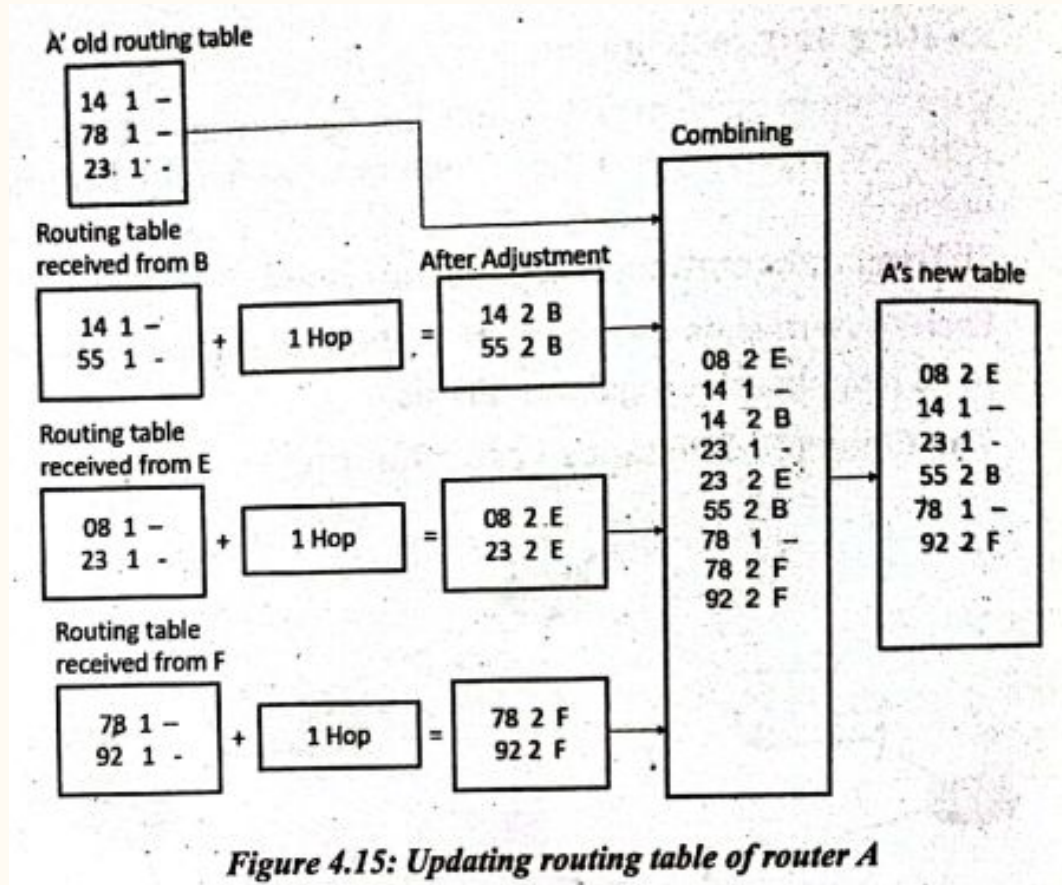
A router maintains a vector (the next hop) and a distance (the metric cost) to all reachable destinations. This process relies on periodic advertisement and local updates, commonly governed by the Bellman-Ford algorithm.

- Routers do not possess a map of the entire network. Instead, each router broadcasts its entire routing table exclusively to its immediate neighbors.
- This creates a "routing by rumor" environment where information is transitive—a router derives its knowledge from what its neighbor claims to know.
- When a router receives a table from a neighbor, it computes the cost to each destination by summing the cost of the link to that neighbor with the cost advertised by the neighbor for the destination.
- If the result provides a lower cost than the current route, the routing table is updated.

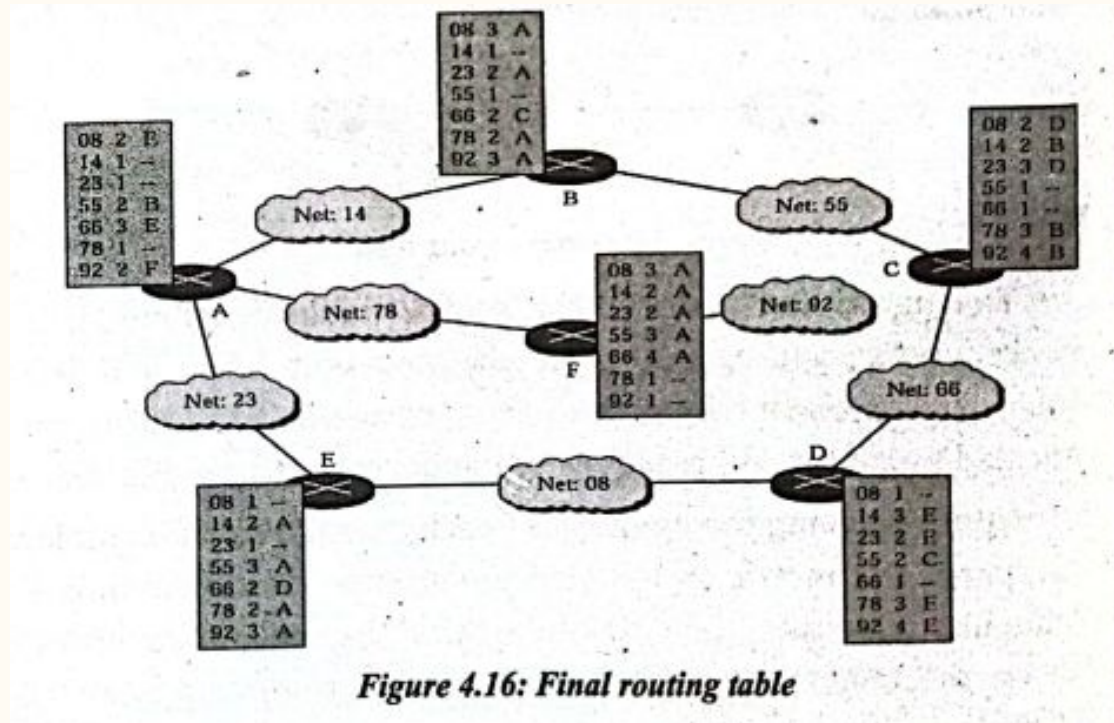
Routing Algorithms: Distance Vector



Routing Algorithms: Distance Vector



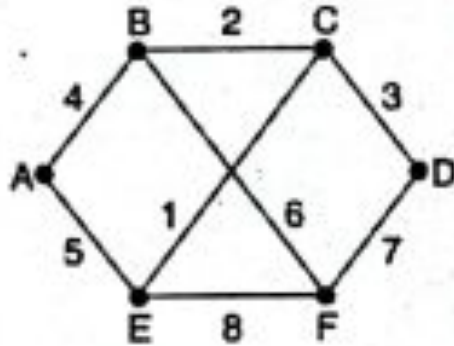
Routing Algorithms: Distance Vector



Routing Algorithms: Link State Routing

- Every router sends a "Hello" packet to its neighbors. This confirms who is connected to whom and if the link is actually working.
- Each router creates a small "status report" called a Link State Advertisement (LSA). This contains just its own local connections.
- The router "floods" this status report to every other router in the network—not just neighbors. Eventually, every single router in the network has a copy of everyone else's status report.
- Every router takes all these reports and builds a master map (database) of the entire network topology.
- Now, every router runs Dijkstra's algorithm on its map. It places itself at the center and calculates the absolute fastest tree-path to every other node.

Routing Algorithms: Link State Routing



Link		State		Packets	
A		B		C	
Seq.		Seq.		Seq.	
Age		Age		Age	
B	4	A	4	B	2
E	5	C	2	D	3
		F	6	E	1

D		E		F	
Seq.		Seq.		Seq.	
Age		Age		Age	
C	3	A	5	B	6
F	7	C	1	D	7
		F	8	E	8

Figure 4.18: A subnet and LSP (link state packets) for this subnet

Internet Control Protocol: ARP

ARP(Address Resolution Protocol) is the bridge between the Network Layer (IP) and the Data Link Layer (MAC). Since Ethernet switches/NICs operate exclusively on MAC addresses, IP packets must be mapped to physical addresses before frame encapsulation.

- When a host (Source A) needs to communicate with a target host (Target B) on the same local network, it performs an ARP lookup.
- **Process:**
 - *ARP Request:* A broadcast frame containing the target IP address is sent to the entire local segment (FF:FF:FF:FF:FF:FF).
 - *ARP Reply:* The target host, recognizing its own IP address, unicasts a reply containing its physical MAC address.
- **The ARP Cache:** To avoid repeating this broadcast for every packet, hosts maintain an ARP Cache (a temporary table) mapping recently resolved IP-to-MAC associations.

Internet Control Protocol: RARP

RARP(Reverse ARP) is a legacy protocol that historically served a specific niche for "diskless" workstations.

Device often knows its own physical MAC address (burned into its NIC) but has no storage to hold an IP configuration. RARP allows such a device to query a centralized server to retrieve an IP address.

RARP is essentially obsolete. It has been replaced by more robust, application-layer protocols like BOOTP (Bootstrap Protocol) and eventually DHCP (Dynamic Host Configuration Protocol), which provide far more comprehensive configuration (subnet masks, gateways, DNS servers).

Internet Control Protocol: ICMP

ICMP(Internet Control Message Protocol) is the diagnostic and error-reporting mechanism of the IP suite. Unlike TCP or UDP, it does not carry user data; it is encapsulated directly in IP packets and serves as the network's feedback loop.

- If a router cannot deliver a packet (e.g., Destination Unreachable), it sends an ICMP message back to the source to explain the failure.
- The ubiquitous ping and traceroute utilities rely on ICMP "Echo Request" and "Echo Reply" messages to verify path reachability and latency.
- ICMP Source Quench messages (deprecated but historically used) were meant to signal senders to reduce data flow when congestion was detected.
- ICMP messages are categorized by Type and Code. For instance, an ICMP Type 3, Code 1 indicates "Destination Host Unreachable."

Routing Protocol

Routing protocols are the automated systems that enable routers to share reachability information. They are divided into two main categories: **IGPs (Interior Gateway Protocols)** and **EGPs (Exterior Gateway Protocols)**.

Before the classification of routing protocols, we need to know about the **Autonomous system (AS)**. An Autonomous system is a group of networks and routers under the authority of a single administration.

Routing inside an autonomous system is referred to as **interior routing**. Routing between autonomous systems is referred to as **exterior routing**.

Routing Protocol: OSPF (Open Shortest Path First)

OSPF is an IGP. It manages routing exclusively within a single administrative domain or Autonomous System (AS), such as your company's internal network.

OSPF utilizes a Link-State algorithm. Unlike Distance Vector protocols that pass summarized information to immediate neighbors, OSPF routers flood Link State Advertisements (LSAs) to every router within an area.

Every router maintains a synchronized Link State Database (LSDB), which serves as a complete topological map of the local network. Routers then run the Dijkstra (SPF) Algorithm on this map to calculate a loop-free, shortest-path tree with themselves at the center.

To manage memory and CPU overhead in large networks, OSPF uses Hierarchical Areas. Area 0 (the Backbone) acts as the central transit hub, ensuring that flooding is confined to specific boundaries.

Routing Protocol: BGP (Border Gateway Protocol)

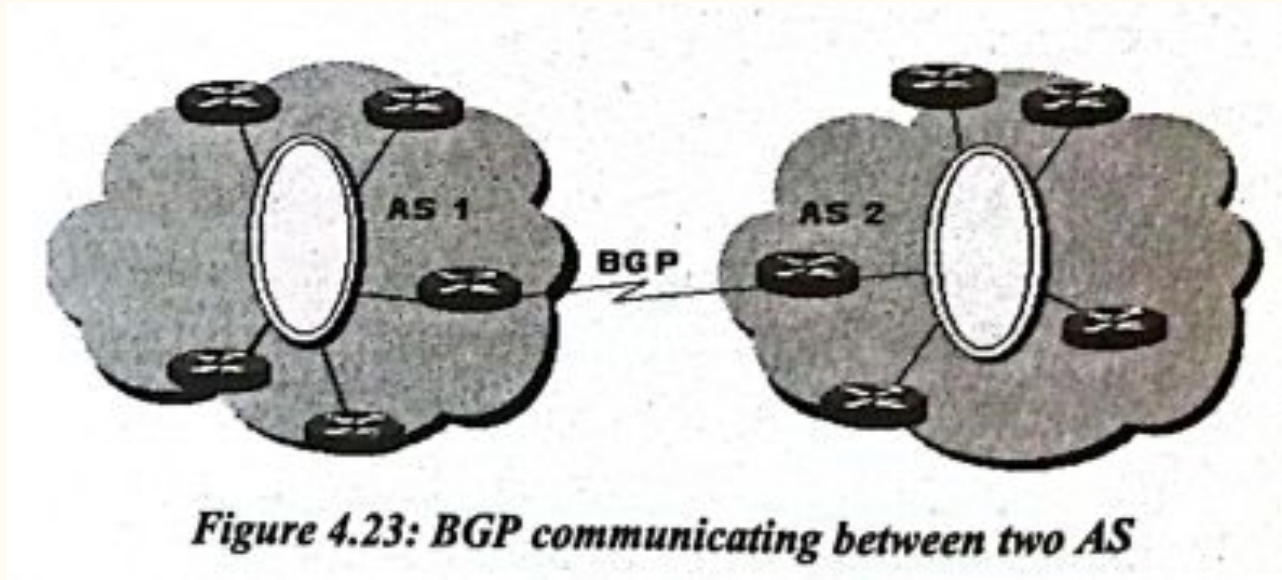
BGP is an EGP. It is the primary protocol of the global Internet, responsible for routing between different Autonomous Systems (e.g., connecting an ISP to a large enterprise or another ISP).

BGP is a Path Vector protocol. It does not prioritize the "fastest" path based on link bandwidth or latency. Instead, it makes routing decisions based on Path Attributes and Administrative Policies.

BGP routers (peers) exchange full path information, including the specific sequence of AS numbers (AS_PATH) a packet must traverse. This allows network administrators to enforce business rules—for example, preferring a specific transit provider for cost reasons, even if it is not the shortest technical path.

Unlike IGPs, BGP operates over TCP port 179. This ensures the reliable delivery of massive, complex routing tables across unreliable global connections.

Routing Protocol: BGP (Border Gateway Protocol)



Traffic Delivery Paradigms

1. Unicast Routing (One-to-One)

Unicast is the most common form of communication on the internet. A single packet is sent from one sender to one specific recipient, identified by a unique destination IP address.

The network infrastructure, including switches and routers, treats each packet as a private conversation. The source address and destination address are explicitly defined in the packet header, ensuring the data follows a dedicated path to the intended host.

This is the foundation for standard web browsing, email communication, and file transfers, where a private, direct connection between two devices is required.

Traffic Delivery Paradigms

2. Broadcast Routing (One-to-All):

Broadcast communication involves sending a packet from one source to every device located within the same local network segment or "broadcast domain."

The sender uses a special broadcast address (such as 255.255.255.255 in IPv4). When a switch receives this packet, it copies the frame to every single port. Crucially, routers act as a firewall for this traffic; they are programmed to discard broadcast packets to prevent "broadcast storms," which would otherwise consume all available bandwidth across the wider network.

This is primarily used for network discovery, such as the Address Resolution Protocol (ARP), where a device needs to find the hardware (MAC) address of a neighbor and must "shout" to the entire segment to get an answer.

Traffic Delivery Paradigms

3. Multicast Routing (One-to-Many):

Multicast is a highly efficient method designed to send a single stream of data to a specific group of recipients who have "subscribed" to a multicast address.

Unlike broadcast, which forces every device to process the packet, multicast only delivers traffic to those that have explicitly requested it. Intermediate switches and routers use sophisticated protocols (like IGMP) to track who wants the data. The network infrastructure replicates the packets only at the points where the path splits toward different interested receivers.

This is the standard for high-bandwidth, real-time applications like IPTV, video conferencing, and financial data feeds, where sending a separate unicast stream to every individual user would quickly overwhelm the network's capacity.

The END